6.1 Area

Name _____ Period ____ Date _____

AP Calculus 6.1 · Student sheet

I can…

- partition an interval into subintervals of equal width.
- approximate an area with left, right, and midpoint rectangles.
- explain how lower and upper sums trap the exact area.

Vocabulary · fill in the missing word

Partition — dividing [a, b] into n _____, each of width $\Delta x = \frac{b - a}{n}$

Lower sum — the sum of rectangle areas using the minimum value of f on each _____

Upper sum — the sum of rectangle areas using the maximum value of f on each _____

Part 1 · Count the rectangles

$y = x^2$ on [0, 2] with 4 rectangles of width 1/2.

Method	Heights	Sum of areas
Left endpoints		
Right endpoints		
Exact area (we'll learn it)	—	8/3 ≈ 2.67

Part 2 · Notes and examples (fill in each note, then work the example)

Partition: n subintervals of [a, b] each have width Δx = _____.

Example 1

Partition [1, 5] into 4 subintervals of equal width. Find Δx and the subintervals.

Lower and upper sums: a lower sum uses the _____ value of f on each subinterval; an upper sum uses the maximum.

Example 2

$f(x) = x^2$ on [0, 2] with n = 4. Find the lower sum and the upper sum.

Midpoints: the midpoint sum uses f at the _____ of each subinterval.

Example 3

$f(x) = x^2$ on [0, 2] with n = 4. Use midpoints.

Exact area: the area is the _____ of the sums as $n \to \infty$; for straight-sided regions, use geometry.

Example 4

Find the exact area under $f(x) = 2x + 1$ from $x = 0$ to $x = 3$.

Part 3 · Partner practice

Partner 1

$f(x) = x + 1$ on $[0, 4]$ with $n = 4$. Use right endpoints.

Partner 2

(AP® Practice) The area under $f(x) = x^2 + 1$ from $x = -1$ to $x = 3$ is approximated with a lower sum using 4 subintervals of equal width. The approximation is

(A) 8 (B) 9 (C) 10 (D) 18